

Predicting Customer Churn and Retention Time: A Study of Four Modelling Approaches

Group 14: Herdify

Abishek Alagan, Aidan Meazzo, Alastair Kirkwood, Hanbo Yu, Harry
Chandrakumar.

May 3, 2026

Mathematical and Data Modelling 3
School of Engineering Mathematics and Technology
University of Bristol



Declaration of AI use

Chat GPT used for code bug fixing and used for spelling and grammar check throughout the project.

1 Introduction

Customer churn, the event when a customer chooses to leave or stop purchasing from a company, is a serious issue across industries especially in sectors like retail. In a 2023 customer retention benchmark report by MoEngage it was found that 45% of the retail brands are now experiencing customer churn rates of 51% or higher over the course of a year, while 18% see 51-60% churn and 14% report a churn rate between 61-70% [1]. These high churn rates result in lost revenue and a reduced ability to maintain a stable customer base for businesses. Accurately predicting when different types of customers are likely to churn can enable businesses to make informed decisions on customer strategies. Furthermore, the ability to estimate the time to churn enables businesses to engage the customers at risk. Studies have shown that acquiring a new customer can cost five times more than retaining an existing one, making customer retention vital for a company [2]. Furthermore, increasing customer retention rates by just 5% can boost profits by 25% to 95% [3]. The results of our project will benefit businesses seeking to optimise their retention strategies.

Industry experts are increasingly recognising the importance of churn prediction done. Early methods used statistical based models such as survival analysis and logistic regression to understand churn patterns and estimate the risk [4]. Additionally survival models are specifically designed to handle time-to-event data, making them highly relevant for estimating the time until churn [5]. Recent advancements see the introduction of machine learning into prediction models such as neural networks, along with hybrid models combining probabilistic and machine learning techniques, like the BG/NBD model [6]. This project builds on these different approaches and evaluates the applicability of them to the data we are provided.

The two primary goals of this project:

1. To predict if a customer has churned based on different churn threshold values.
2. To estimate the time until a customer churns for customers who have not yet churned.

This report investigates the process of predicting customer churn, beginning with a data overview section including data preparation, feature engineering and an exploratory data analysis. Overall, this project provides insights for the implementation of customer retention strategies through the modelling of customer churn.

2 Dataset Overview

The datasets provided by Herdify included key details on each customer’s purchasing behaviour: total number of purchases, total spending, date of first purchase and date of last purchase.

Four datasets were provided, each relating to distinct business sectors: two food retailers, a charity and a financial service. The datasets concerning the food retailers was selected to train, analyse and evaluate our models against due to the large number of customer data points and high variation in the total number of purchases compared to the other two datasets, making it suitable for estimating customer churn. The financial service data was selected for stress testing the models as it has a low number of purchases per customer but high purchase value.

To prepare the data for our models, it was first cleaned. Customers with a negative or zero total spending — potentially due to giveaways or refunds — were removed as their data would skew the accuracy of our models. The first and last purchase dates were converted into date-time format to allow accurate time-based calculations.

Finally feature engineering was completed on the data to enhance the analysis. Columns for average purchase price, tenure and purchase frequency, were all generated.

2.1 Exploratory Data Analysis

The second dataset is analysed because it is used for training the models. 2D Principal Component Analysis (PCA) technique analyses all the features simultaneously. PCA is a technique that incorporates a change in a multi-featured dataset into two principal components (PCs) via dimensional reduction. The two PCs are then plotted on a scatter graph to allow for any sort of visualisation of patterns. Before using the PCA method, the features are normalised to prevent any sort of bias towards a much greater value compared to a value of smaller magnitude [7] such as the total value of a customer's purchase and the frequency of a customer's purchase.

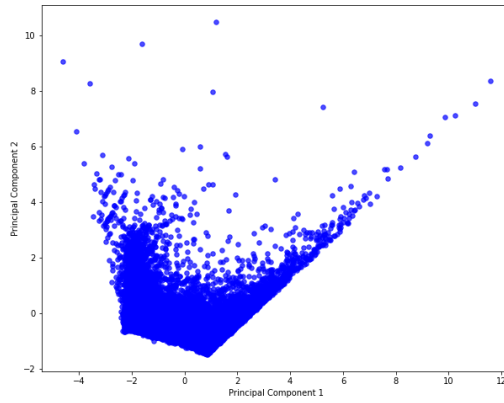


Figure 1: PCA graph of First Dataset With No Colour Gradient

Figure 1 shows a high concentration of data points near the bottom of the graph suggesting common customer behaviour. There are also outliers to indicate extremities due to certain features which could determine any high value or low value customers.

The next approach is to use the eigenvalues of each PC to find the variance of the data covered by the PCs [8]. The eigenvalues for PC1 and PC2, respectively, are 2.7388 and 1.1280. From this, it can be said that PC1 covers 70.8% of the overall dataset while PC2 covers 29.2%. To see which features have the greatest impact on the PCs, finding the eigenvectors gives an idea of the impact each feature. Each element in the eigenvectors is known as the loading element which indicates how strongly a feature influences a PC. The first element is the loading contribution towards PC1 and the second is towards PC2 [8].

Figures 23 and 24 show the loadings of all the features towards both PCs. The number of purchases and the total value of a customer's purchase have the greatest effect on PC1 and for PC2, the average purchase price and the days since first purchase have the greatest impact.

To verify the eigenvectors, a colour gradient can be used. In this type of graph, a colour gradient varies with the value of the features to identify patterns. Each graph plotted has a colour gradient in the direction of the different eigenvectors which proves that the eigenvector method works for the features and can find how useful each feature is to the PCs (see Appendix C for further detail).

To further improve the PCA method, 3D PCA is used to reduce the loss in data that occurs in the 2D PCA. The 3D visualisation can help see any relationship from a spatial view. When the PCA is redone in a three-dimensional space, the eigenvalues for PC1, PC2 and PC3, respectively, are 2.7388, 1.1280 and 0.9649. Even though PC3 has a lower eigenvalue than PC1 and PC2, it still covers 20% of the variance of the data, meaning it does capture a reasonable portion of the dataset’s structure. The eigenvector values show that the frequency of a customer’s purchases (loading contribution of 0.822945) and the days since the first purchase (loading contribution of 0.536037) have the greatest impact on PC3.

Overall, PC1 seems to represent more of a customer’s spending volume, PC2 seems to denote the loyalty of a customer and PC3 reflects the engagement of a customer. This information helps find out how long for a customer to churn.

3 Logistic Regression

Logistic regression is commonly used for binary classification tasks, making it suitable for predicting customer churn. Logistic regression provides a straightforward and interpretable model for predicting customer churn and how each feature influences churn likelihood, which is particularly valuable for stakeholders seeking insights into customer behaviour [9].

Logistic regression is used to predict churn by analysing five key features — Frequency, Average Purchase Price, Total Spend, Tenure, and Number of Purchases — each representing important aspects of customer behaviour. Logistic regression is a statistical method that models the probability of a binary outcome, which in this case is whether a customer has churned or not. This is done by fitting a linear equation to the data and then applying a logistic function to map the output to a probability between 0 and 1. To make the final classification, a threshold of 0.5 is applied, assigning 1 for churn if the probability exceeds the threshold and 0 if it does not. This model also includes an L1 penalty (Lasso regularization), which shrinks the coefficients of less important features. This means the model focuses on the most impactful predictors and discards less relevant ones reducing overfitting.

The model is trained and validated across different churn thresholds (50, 100, ..., 500 days), with each iteration completing 10 repetition to get an average accuracy using the first food retailer’s dataset. The trained model is then evaluated using the second food retailer’s dataset to find the accuracy of the model again with 10 repetitions.

3.1 Analysis of Results

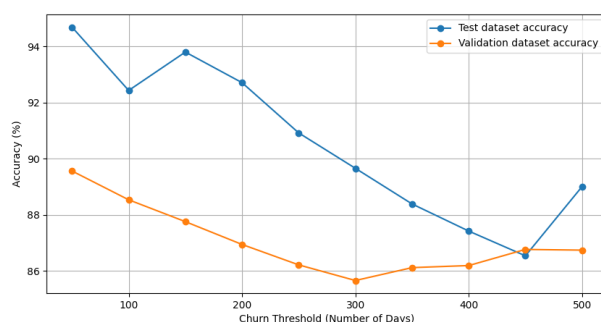


Figure 2: Logistic regression’s accuracy against churn thresholds

Figure 2 shows the accuracy of the logistic regression model across different churn thresholds. The test dataset accuracy consistently outperforms the validation dataset accuracy across all thresholds, suggesting that it has patterns that the model is capturing well because of sparsity or variability in the test dataset. At lower churn thresholds, the model achieves high accuracy on the test dataset, with values above 94% and an accuracy of 90% to 86% for the validation dataset. Both accuracies show a noticeable decline as the churn threshold increases. This pattern shows that the logistic regression model is sensitive to the threshold definition, indicating that as the time window for churn extends, it becomes harder for the model to capture specific patterns in customer behaviour because the patterns become less linear, reducing its predictive performance. Overall, the logistic regression model displays a high accuracy for low churn thresholds based on the dataset features but reveals limitations in adapting to higher churn thresholds. For these lower churn thresholds, it is a viable option for business use.

The python time library was used to calculate the average run time for each churn threshold which was 0.61 seconds. This is a very efficient processing time considering the model is being trained and validated on a dataset of 180 thousand customers and tested on a dataset of 1.8 million customers. The quick runtime highlights the computational efficiency of logistic regression, making it a suitable choice for large customer churn prediction tasks where fast results are essential.

In figure 3, the churn threshold was set to 250 and the model was then evaluated excluding different features. Removing tenure and number of purchases leads to the most significant drops in accuracy, indicating these are essential predictors for churn while average purchasing price and total spend are not. Unexpectedly, excluding frequency yields a higher accuracy of 93.03% than when it's included seen in 2, suggesting that while frequency is useful it may sometimes add noise at this threshold. Further analysis of removing frequency from the main model reveals that it significantly reduces the performance, with the test accuracy dropping more at higher churn thresholds and showing high variability across churn thresholds shown in figure 26. This suggests frequency is essential for stable predictions.

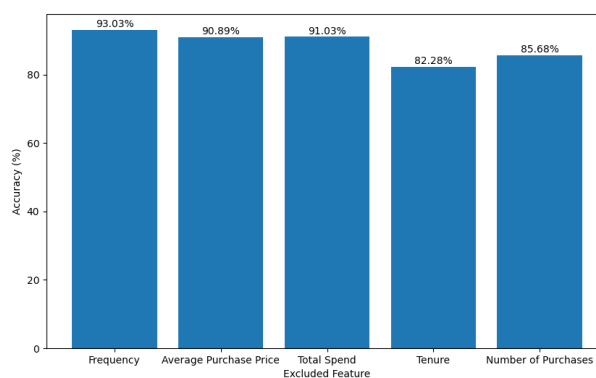


Figure 3: Bar graph showing how excluding each feature affects the accuracy of the logistic regression model with a churn threshold of 250 days

To stress test the model different training set sizes, 1%, 10%, and 100% of the original data are used on the model. Figure 4 displays the test accuracy of the logistic regression model across varying churn thresholds. It reveals that across all training set sizes, the accuracy patterns are relatively consistent, proving that the logistic regression model is robust and can generalize well

even when using significantly reduced data for training it.

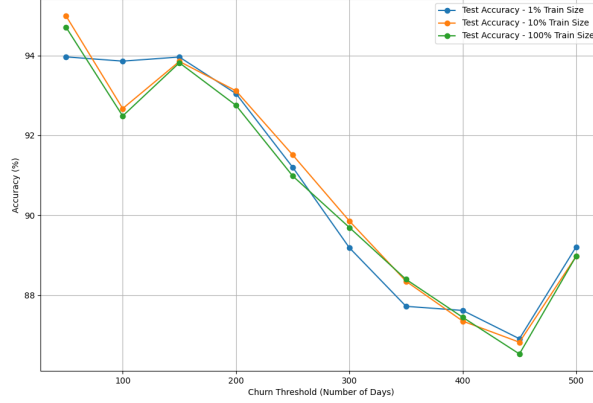


Figure 4: Logistic regression’s accuracy against churn thresholds with varying sizes of training data

The evaluation test dataset was also changed to the finance dataset to see if there was any accuracy differences because the dataset had very different ranges in total spend and total number of purchases. However the accuracy stayed very high and similar to before, maybe even slightly higher as seen in figure 27.

4 Neural Network

Neural networks are suited to predicting if a customer has churned for complex data due to their ability to identify non-linear relationships in customer behaviour data. It can act as a binary classifier and assign them a 0 or 1 value depending on whether they have churned or not. Unlike other methods, neural networks can recognize complex patterns across multiple different features. This versatility makes them effective in identifying very small signals of churn that simpler models would miss and adjusting to different churn threshold definitions. Studies have shown that neural networks achieve higher accuracy rates than traditional models such as logistic regression or k-nearest neighbours due to their ability to capture these complex non-linear patterns [10].

The same five key features are used as inputs to the models to achieve the first primary objective. The neural network processes these inputs through a series of interconnected layers. Each layer contains neurons, which perform calculations by applying weights to inputs and passing the results through a final activation function. The model predicts whether a customer has churned or not and calculates the accuracy depending on the varied churn threshold.

The neural network’s architecture (see figure 10 for a diagram):

1. Input Layer: This layer processes the five features.
2. Hidden Layers: The model includes 1 hidden layer containing 8 neurons, using the ReLU activation function to capture the complex patterns in customer behaviour. This parameter was decided by comparing the models computational load against the results it outputted see figures 11, 12, 13, 14, 15 for further reasoning.

3. Output Layer: A single neuron with a sigmoid activation outputs a value of 0 or 1, indicating if they have not churn or have churn respectively.

The model is trained and validated across the same churn thresholds as before, with each iteration completing 10 epochs this time. With each epoch, the model improves its understanding of the patterns in the data, improving its accuracy and reducing its loss. Ten epochs proved enough as the model’s accuracy plateaued after reaching this, showing it had learned from the data. Additional epochs would be unnecessary and would lead to overfitting and an increase in computational time, without giving improvements in the models performance. The trained model is then evaluated.

4.1 Analysis of Results

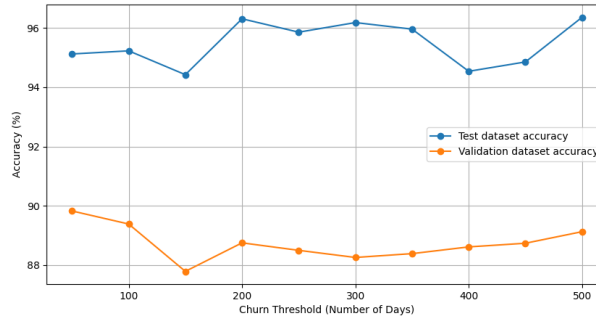


Figure 5: Neural network’s accuracy against churn thresholds

Figure 5 presents the neural network’s accuracy on the test and validation datasets across various churn thresholds, ranging from 50 to 500 days. The test dataset accuracy is always higher than the validation dataset accuracy, with it varying between 94% and 97%. This suggests the model is learning the complex patterns in the data effectively to be able to maintain the high levels of accuracy, but there is a notable difference between the test and validation accuracies. The validation accuracy remains around 88% to 90%, with a small amount of fluctuation in the lower churn thresholds indicating a bit of overfitting. The difference between the two suggests that the model has learned patterns that are specific to the test data and can be sensitive to changes in churn thresholds, while the validation accuracy reflects a more generalisable performance. Despite the difference, the neural network’s ability to maintain high accuracies across various churn thresholds suggests its effectiveness in modelling customer churn. This shows that the neural network is perfect for tasks requiring identifying non-linear relationships. This robustness highlights its potential for real-world customer retention applications.

The average run time for this model for each churn threshold was 31.45 seconds. This is a slow processing time caused by the large size of the training, validation and testing datasets. This shows that the neural network isn’t as computationally efficient for running extremely large datasets.

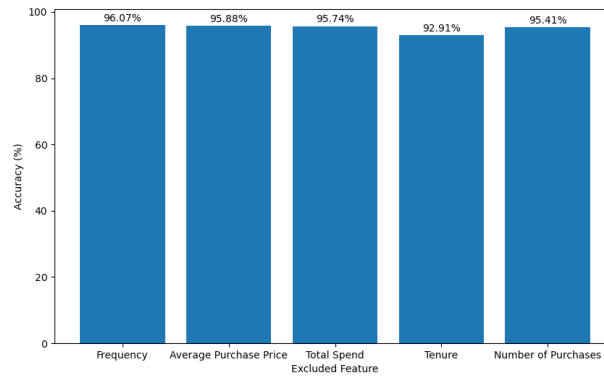


Figure 6: Bar graph showing how excluding each feature affects the accuracy of the neural network model with a churn threshold of 250 days

In figure 6, the churn threshold was set to 250 and the model was evaluated excluding different features. Only tenure leads to a small drop in accuracy, indicating it is an essential predictor for churn while the rest not so much.

To stress test the model, different training set sizes of the original data were used to assess the neural network's performance. Figure 7 displays the test accuracy of the model across varying churn thresholds, showing a clear correlation between training dataset size and accuracy. As the training size decreases, the model's accuracy drops, highlighting its dependency on larger datasets to capture complex patterns in customer churn.

The evaluation test dataset again was changed to the finance dataset to see if there was any accuracy differences because the dataset had very different ranges in total spend and total number of purchases. However, the accuracy stayed very high and similar to before, maybe even slightly higher as seen in figure 28.

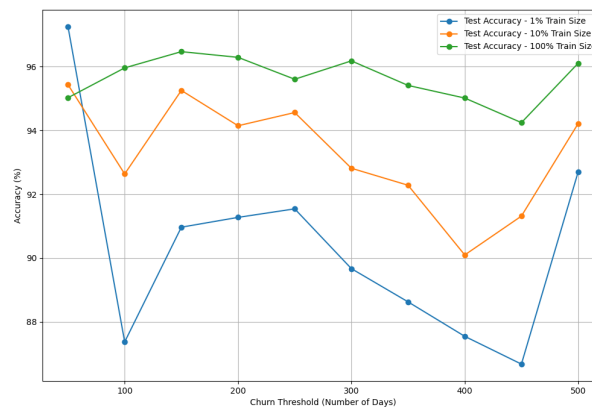


Figure 7: Neural Network's accuracy against churn thresholds with varying sizes of training data

5 Survival Model

Survival models consist of statistical methods, used to track time until an event occurs. In Customer Lifetime Value (CLV) models, the event we are interested in is customer churn which allows us to model the likelihood of a customer staying active over specified time periods, based on previous behaviour. This survival model uses the Cox Proportional Hazards model (CoxPH) because it is semi-parametric and does not require any assumptions about the baseline hazard model, making it more flexible for CLV models. It models the hazard (risk of churn) rate as a function of time and covariates. The covariates that applies to the data set are: recency (number of days since last purchase); frequency (number of transactions made during the observed period of time); monetary value (total value of customers transactions).

The hazard function used in the CoxPH model is as follows:

$$h_{t|X} = h_0(t) \times \exp(\beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3) \quad (1)$$

Where $h_{t|X}$ is the hazard rate at time t with covariate values X . $h_0(t)$ is the baseline hazard rate when all covariate values are 0. β s are the coefficients of each covariate that represents its effect on the hazard rate. X s are the covariate values.

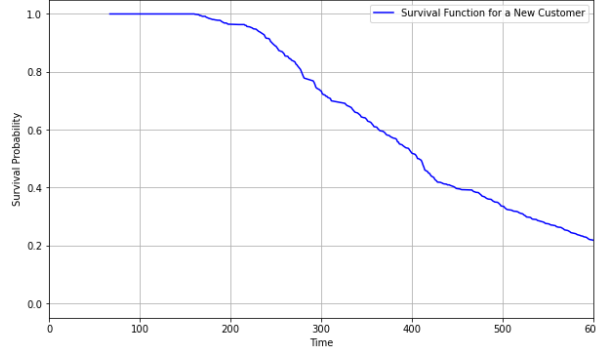


Figure 8: Survival Function

Figure 8 displays the probability that a customer is still active against time in days and helps to visualise the risk of customer churn over specified time periods. The curve shows that, overall, the risk of churn remains relatively small for the first 200 days, after which there is a steady decline.

5.1 Analysis of Results

Since a survival model generally analyses the 'time to event', it is more suited for predicting the risk of churn over time as opposed to being a direct binary classifier. As a result, this model cannot be used to predict whether or not a customer has churned. However, it is possible to predict the time until a customer churns by using when the survival rate of a customer reaches 50% which represents the median time to churn.

This model returns a coefficient for each covariate under the CoxPH model which tell us how and to what extent each covariate affects the hazard of churn. Recency and frequency both have a positive coefficient of 0.02 and 0.06 respectively which suggests the hazard of churn increases with these covariates. This makes sense for recency as customers with the higher number of days since their last purchase are more likely to have churned. The frequency covariate may seem counter-intuitive as it suggests that customers who purchase more frequently are at a higher risk

of churn. This may imply that there is an issue with the quality of the product and customer satisfaction is low but this interpretation requires specific context and data distribution to corroborate. Monetary value has a coefficient -0.01 which implies customers who spend more money are at a lower risk of churning.

6 BG Model

The Beta-Geometric/Negative Binomial Distribution (BG/NBD) is a probabilistic CLV model that uses non-contractual, continuous time transactions from customers to predict future buying behaviour and customer churn. This model was developed by Fader, Hardie, and Lee [11] as an alternative to the already existing Pareto/Negative Binomial Distribution (Pareto/NBD). Rather than relying on assumptions that each customer has identical purchasing patterns, the BG/NBD model, referred to as BG hereafter, assumes that individual customers exhibit unique behaviour patterns, more specifically we follow the assumptions outlined in [11] listed below:

1. While active, the number of transactions made by a customer follows a Poisson process with transaction rate λ

$$f(t_j | t_{j-1}; \lambda) = \lambda e^{-\lambda(t_j - t_{j-1})}, \quad t_j > t_{j-1} \geq 0$$

2. Heterogeneity in λ follows a gamma distribution with pdf

$$f(\lambda | r, \alpha) = \frac{\alpha^r \lambda^{r-1} e^{-\lambda\alpha}}{\Gamma(r)} \quad (2)$$

where r is the shape parameter and α is the scale parameter for the purchasing process.

3. After any transaction, a customer becomes inactive with probability p . Therefore the point at which the customer “drops out” is distributed across transactions according to a (shifted) geometric distribution with pmf

$$P(J) = p(1 - p)^{j-1}, \quad j = 1, 2, 3, \dots$$

where J is the assumption a customer becomes inactive immediately after the j th transaction

4. Heterogeneity in p follows a beta distribution with pdf

$$f(p | a, b) = \frac{p^{a-1}(1 - p)^{b-1}}{B(a, b)} \quad (3)$$

with a and b are both shape parameters for the dropout process, where $B(a, b)$ is the beta function, which can be expressed in terms of gamma functions: $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$.

5. The transaction rate λ and the dropout probability p vary independently across customers.

6.1 Bayesian Statistical Methods

To effectively estimate the model parameters, we employ a Bayesian statistical approach. Bayesian inference provides a structured framework to incorporate both prior beliefs, known as priors, and observed data when estimating model parameters. Rather than calculating a single value as an estimate for a parameter, Bayesian methods calculate a range of possible

values for each parameter, represented as a posterior distribution. This distribution uses the prior knowledge about customer behaviour patterns and the information observed in the data, allowing us to capture the some of the inherent uncertainty in the parameters and make more informed predictions about future customer actions.

Estimating the posterior distributions for the parameters r , α , b and a is done through a No-U-Turn Sampler (NUTS), using the python package ‘pymc_marketing’. NUTS is an extension of the Hamiltonian Monte Carlo method (HMC), which is a Markov chain Monte Carlo algorithm that avoids the random walk behaviour by taking a series of steps informed by first-order gradient information [12]. NUTS uses a recursive algorithm to build a set of likely candidate points that spans a wide swath of the target distribution, stopping automatically when it starts to double back and retrace its steps [12]. In the context of CLV modelling, the parameters r , α , b and a are sampled with NUTS using 4 independent sampling chains sequentially to approximate their posterior distributions, updating the initial prior distributions using the purchasing data of each customer.

The data was changed to an alteration of an RFM format (Recency, Frequency, Monetary value). This new dataset contains columns using the following nomenclature; frequency (represents the number of repeat purchases a customer has made, i.e. one less than the total number of purchases), recency (represents the time period when a customer made their most recent purchase, which is equal to the duration between a customer’s first and last purchase. If a customer has made only 1 purchase, their recency is 0), T (represents how old a customer is, which is the number of days between a customers first purchase and the most recent purchase of all customers).

6.2 Analysis of Results

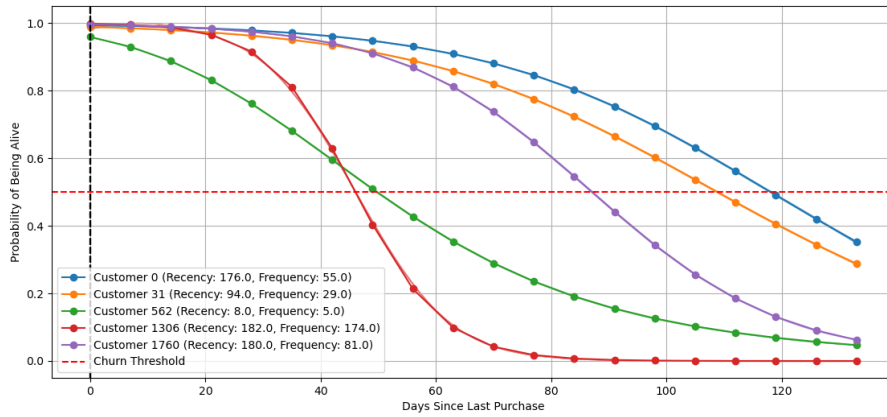


Figure 9: Plot of the mean of $P(\text{alive})$ against time for 5 different customers with a churn threshold of $P(\text{alive}) = 0.5$. The shaded regions around the mean represent the HDI (High Density Interval) probability of 89% around the values of $P(\text{alive})$.

Due to the fact that the BG model is a probabilistic model, it cannot be used as a classifier to determine if a customer has churned after a specific number of days. Instead it can be assumed that a customer has churned once their probability of being alive is less than 0.5 in the case that we have little to no information about the posterior distributions. This threshold could be

lowered to reduce false-positives, but in the absence of information this also increases the risk of creating false-negatives. This model is therefore suited to try and complete primary goal 2.

Figure 9 is a representation of how the values for frequency and recency affect the time it takes for a customer to churn. It can be seen that customer 1306, coloured in red, has a much more severe decrease in $P(\text{alive})$ relative to the other presented in the plot. It therefore isn't a poor assumption that the value of the frequency has a significant impact on the rate of churn, especially as these values become large. Customer 562, shown in green, has low values for both frequency and recency, which is also indicated by the initial probability of being alive being lower than that of the other customers. Due to the fact the frequency is low, the recency becomes more influential in the trajectory of $P(\text{alive})$ over the given time interval. This low recency results in a smoother line for the mean of $P(\text{alive})$, resulting in a similar churn time to that of Customer 1306 with the threshold at 0.5.

Plotting graphs like Figure 9 over longer time periods of time shows one of the weaknesses of the BG model, as the HDI becomes extremely skewed at the beginning of the plot for each customer. The worst case is for customers with high frequency and recency values, causing the values within the HDI to be above 1, which is clearly unreasonable. This is likely due to the fact that high frequency customers can produce skewed posterior distributions as the model is estimating a very high probability of being alive for these customers after their most recent purchase. This can push the HDI boundaries to values that logically don't make sense, i.e. values above 1, since there is less uncertainty about the purchasing behaviour around the most recent purchase and after there have been no purchases for a long period of time, creating noise in-between. A visualization can be seen in Section F of the Appendix in Figure 29

7 Comparison and Discussion of Models

The logistic regression and neural network models demonstrated both strengths and limitations in predicting customer churn, though their suitability varied based on the dataset and computational load. Logistic regression, provided consistently high accuracy at lower churn thresholds. This suggests it is effective when churn is expected within shorter time frames, where the patterns in customer behaviour are possibly more linear. The inclusion of L1 regularization strengthened the model, focusing on the most impactful features which prevented overfitting. Despite its relative simplicity, logistic regression achieved solid accuracy with low computational cost, making it well-suited for the datasets provided. In contrast, the neural network model has the capability to capture non-linear relationships in the data, helping to identify subtle patterns in customer behaviour. It achieved high test accuracy across all churn thresholds, indicating better flexibility than logistic regression in adapting to different definitions of churn. However, the model displayed very high sensitivity to dataset size, with its performance being notably affected when trained with smaller datasets whereas the logistic regression model did not. The neural network also produced a much larger computational load taking roughly 60 times longer to run. Overall, logistic regression offers a balance of accuracy and efficiency, ideal for straightforward churn prediction tasks where there are linear relationships in the data, while the neural network model excels in capturing complex behavioural patterns, although at a higher computational cost. These findings show the importance of a correct model selection based on the specific objectives, data complexity, and computational resources available in customer churn prediction.

The CoxPH and BG models produce results as risk of churn over time, so it is appropriate to analyse their applications within the context of customer churn. The CoxPH model doesn't

require priors as one of its inputs, whereas the BG does as these priors are updated to produce the posterior distributions. If not much is known about the data, this can induce some amount of error into the posterior, especially if there is a small amount of data being used to update these priors. The survival model is a highly versatile tool in business analytics, especially when predicting the time to churn. The ability to forecast the time until a customer's next purchase is useful for retail stores when planning inventory and stocks. Understanding the time frame over which spending occurs allows businesses to plan for future ventures based on predicted revenue streams and cash flow. The BG model is best used for predicting future purchases of customers, which isn't the main goal of this report, as evaluating whether or not customers are still active is a larger concern. Regardless of this, future purchases can give some indication of if and when a customer will churn, as customers that are predicted to make many future purchases within a short time period of their most recent purchase are less likely to churn within that time period as they are still active. The survival model has a relatively low run time but it increases with the size of the dataset and the number of covariates. Using larger amounts of data can be problematic for the BG model as it significantly increases runtime, which is a limiting factor as the model gets more accurate measures for posterior distributions when sampling chains with more customers. Running the BG model using all 189626 customers had an average runtime of around 114 minutes, compared to just under 4 minutes when using the first 5000 customers.

To take this project further, a combination model could be explored, combining predictions from Logistic Regression, Neural Networks, and the Survival/BG models for improved accuracy and robustness being able to run all the models together. Additionally, if real time data was available the changes in the models as they adapt could be explored.

8 Conclusion

This project examined four key models for predicting customer churn: Logistic regression, Neural networks, Survival model, and the BG/NBD model, each serving different purposes in line with the project's two primary goals. Logistic regression proved effective for churn prediction at low churn thresholds, delivering high accuracy with a fast computational time, especially suited for applications with linear relationships. In contrast, the Neural Network, while computationally more intensive, was great at identifying non-linear patterns, offering high accuracy across all churn thresholds. This capability showed its potential to capture complex behavioural patterns within customer data, with the downside of requiring larger datasets to avoid overfitting.

The Survival Model (CoxPH) and BG/NBD models provided value by producing results as functions of time using previous customer data. The CoxPH model estimated churn likelihood over time without requiring prior knowledge of the underlying data, whereas the BG/NBD model's probabilistic approach incorporated the variability of each customer's spending habits, accommodating unique purchasing patterns across users. Although more computationally demanding, the BG/NBD model proved useful in predicting purchasing timelines, highlighting its relevance when evaluating lifetime value beyond binary churn prediction.

Overall, the Logistic regression and Neural network models offer approaches for immediate churn classification, while the Survival and BG/NBD models provide deeper temporal understanding, essential for a business's long term retention strategy.

References

- [1] MoEngage. Customer retention benchmark report. *Modern Retail*, 2023.
- [2] Gallo A. The value of keeping the right customers. *Harvard Business Review*, 2014.
- [3] Schefter P Reichheld FF. The economics of e-loyalty. *Harvard Business School*, 2000.
- [4] Nie G et al. Credit card churn forecasting by logistic regression and decision tree. *Expert Systems with Applications*, 38(12):15273–15285, 2011.
- [5] Coombs N Dudley WN, Wickham R. An introduction to survival statistics: Kaplan-meier analysis. *Journal of the advanced practitioner in oncology*, 7(1), 2016.
- [6] Reinartz W Castéran H, Meyer-Waarden L. Modeling customer lifetime value, retention, and churn. *Handbook of Market Research*, pages 1–33, 2017.
- [7] Jaadi Z. Principal component analysis (pca): A step-by-step explanation, 2024. Accessed: 27/10/24.
- [8] Adewumi J. Understanding the role of eigenvectors and eigenvalues in pca dimensionality reduction., 2019. Accessed: 29/10/24.
- [9] Hastak M McCarty JA. Segmentation approaches in data-mining: A comparison of rfm, chaid, and logistic regression. *Journal of Business Research*, 60(6):656–662, 2007.
- [10] Akcayol MA Utku a. Neural network based a comparative analysis for customer churn prediction. *Muş Alparslan Üniversitesi Fen Bilimleri Dergisi*, 2024.
- [11] Hardie BGS Lee LK Fader PS. “Counting Your Customers” the Easy Way: An Alternative to the Pareto/NBD Model. *Marketing Science*, 24(2):275–284, April 2004.
- [12] Gelman A Hoffman MD. The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47):1593–1623, 2014.

A Neural Network

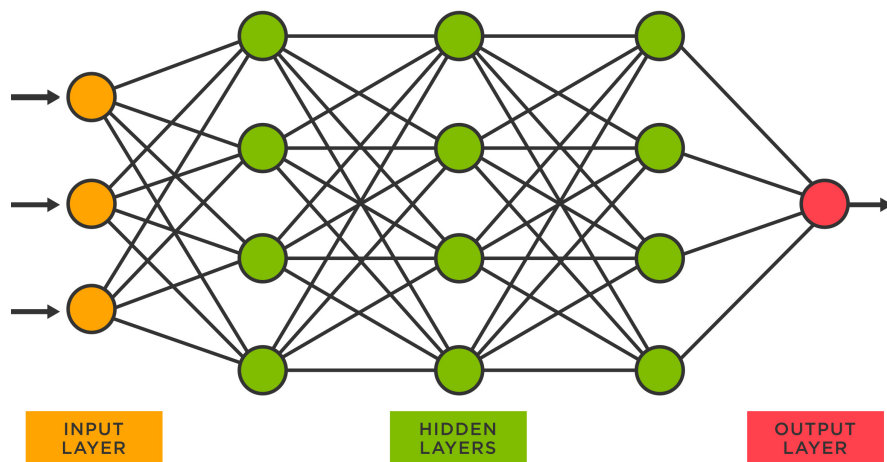


Figure 10: Basic Neural Network Digram

B Different Neural Network Structures

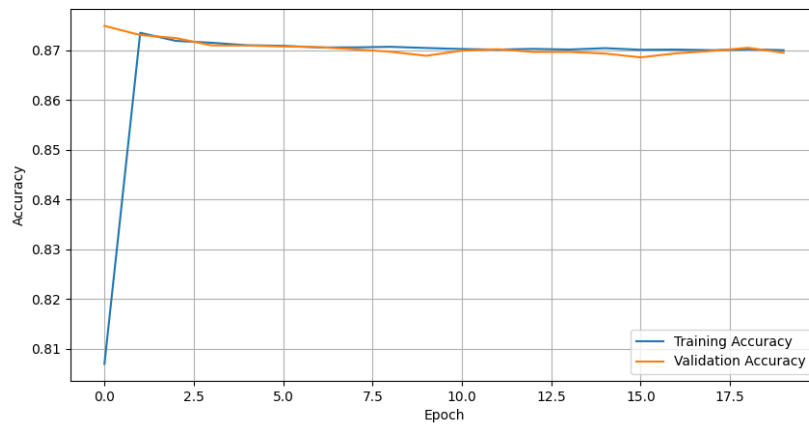


Figure 11: Neural network with 0 hidden layers

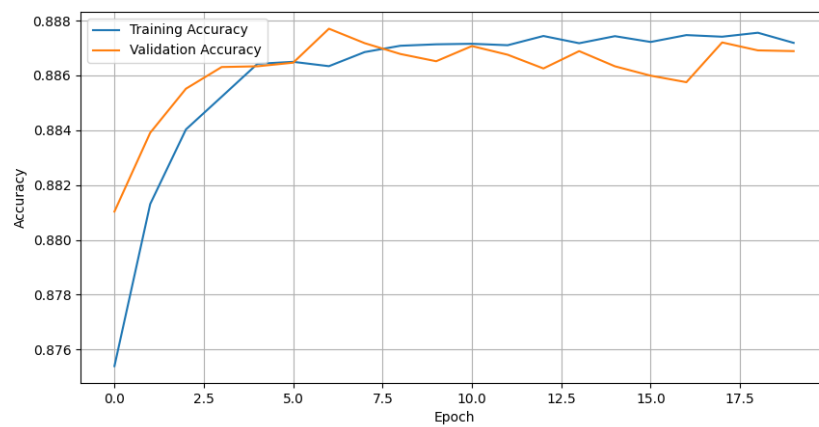


Figure 12: Neural network with 1 hidden layer containing 8 neurons

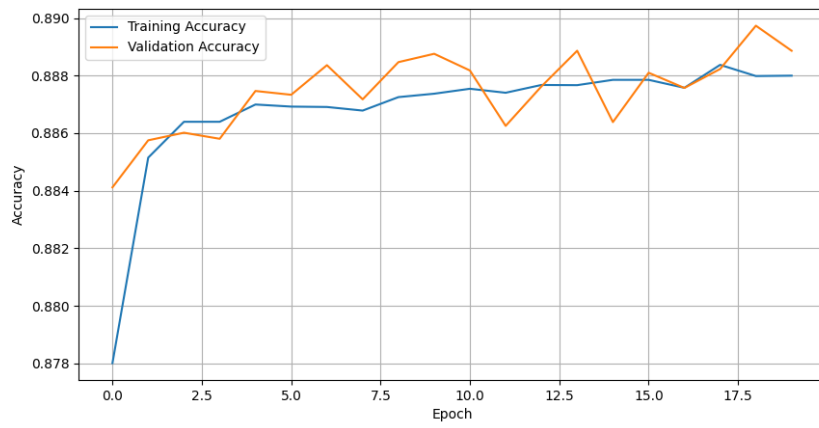


Figure 13: Neural network with 1 hidden layer containing 128 neurons

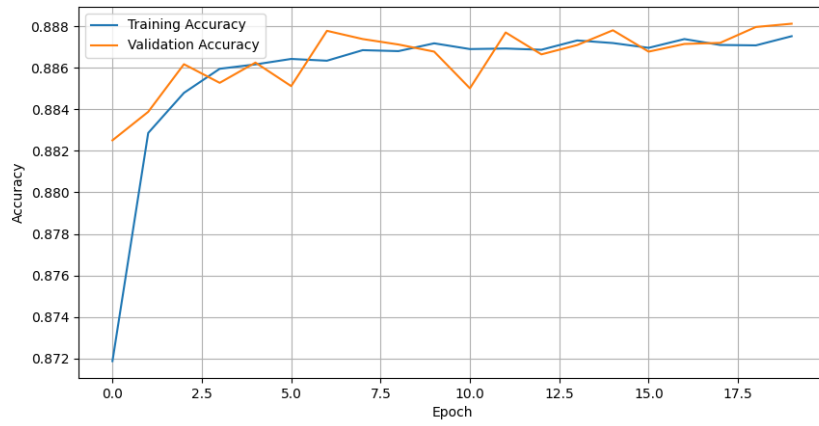


Figure 14: Neural network with 3 hidden layers containing 8 neurons each

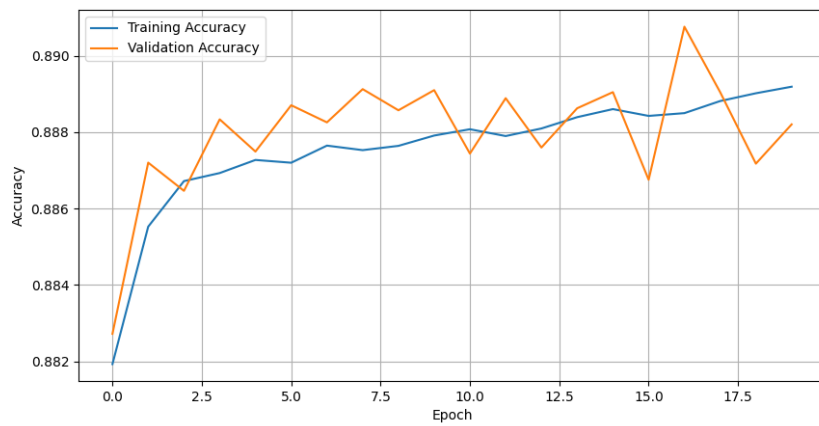


Figure 15: Neural network with 3 hidden layers containing of 128, 64, 16 neurons respectively

C Exploratory Data Analysis

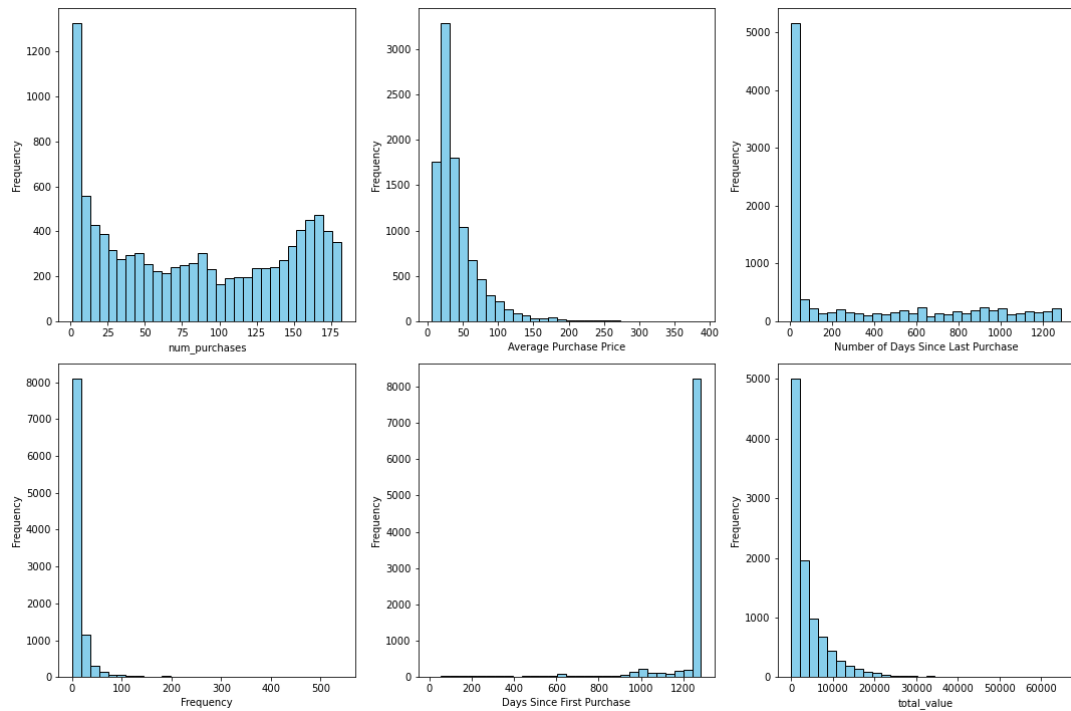


Figure 16: Histogram of all features

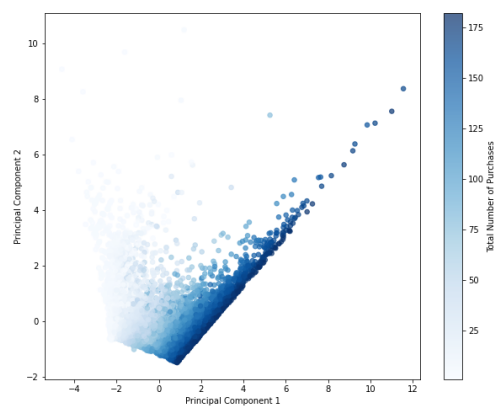


Figure 17: PCA Graph with a colour gradient with respect to the number of purchases

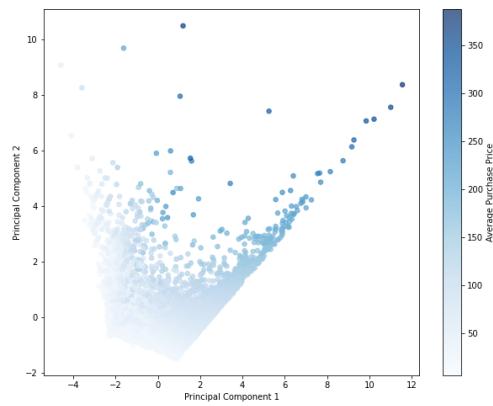


Figure 18: 2D PCA Graph with Colour Gradient using Average Purchase Price

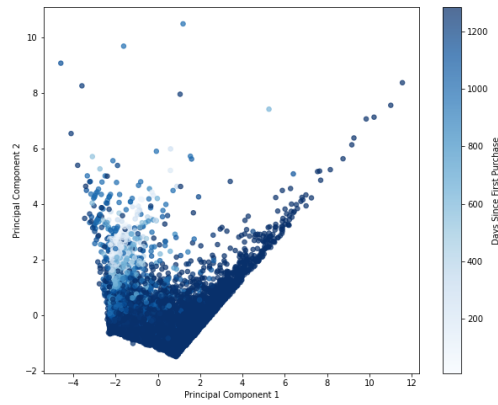


Figure 19: 2D PCA Graph with Colour Gradient Using the Number of Days since First Purchase

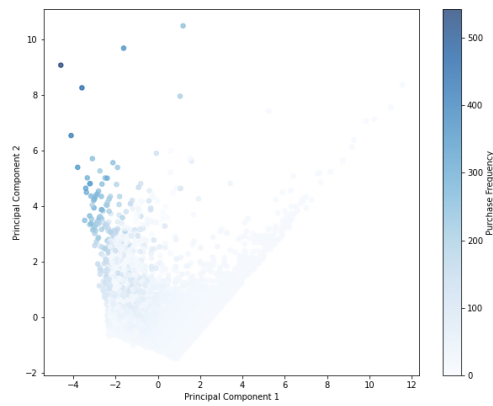


Figure 20: 2D PCA Graph with Colour Gradient Using Frequency of Purchase

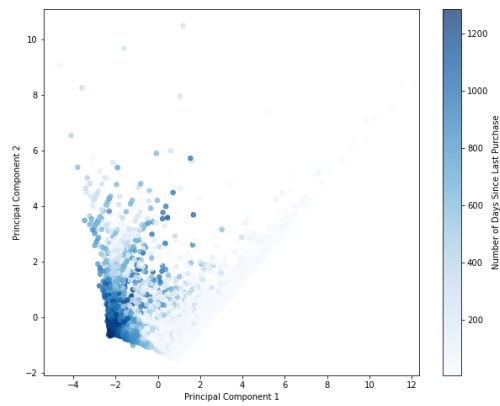


Figure 21: 2D PCA Graph with Colour Gradient Using the Number of Days Since the Last Purchase Made

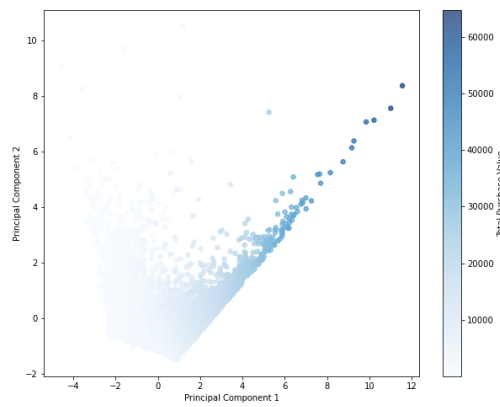


Figure 22: 2D PCA Graph with Colour Gradient Using the Total Value of All the Purchases

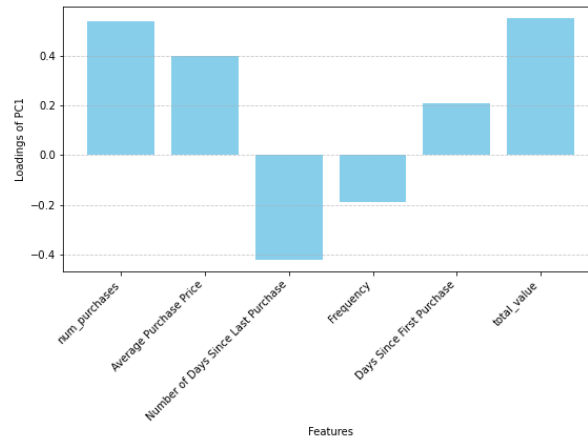


Figure 23: Loading Contributions of Each Feature Towards PC1

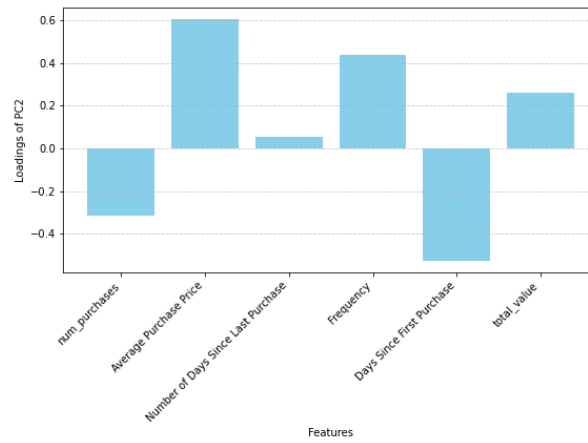


Figure 24: Loading Contributions of Each Feature Towards PC2

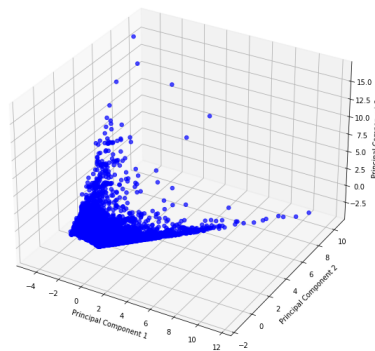


Figure 25: 3D PCA Graph with a colour gradient with respect to the number of purchases

	PC1	PC2	PC3
Number of purchases	0.535110	-0.313633	0.048389
Average purchase price	0.399167	0.603324	-0.068074
Number of Days Since Last Purchase	-0.418448	0.055909	-0.161866
Frequency of Purchase	-0.185534	0.436973	0.822945
Days Since First Purchase	0.209411	-0.526195	0.536037
Total Value of Purchases	0.548588	0.258227	-0.047433

Table 1: Loading contribution of each feature towards each PC

D Testing models excluding frequency as a variable

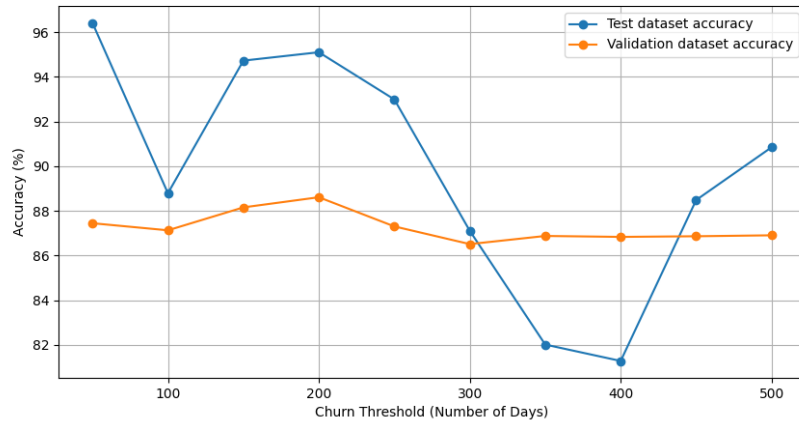


Figure 26: Logistic regression's accuracy against churn thresholds without using frequency as a feature

E Evaluating model on finance dataset

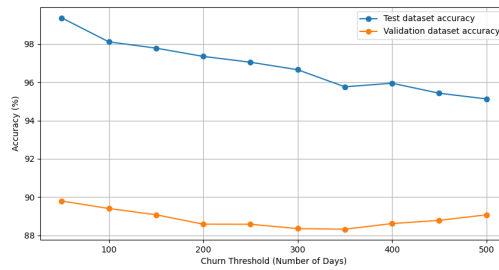


Figure 27: Neural network's accuracy against churn thresholds using the finance dataset

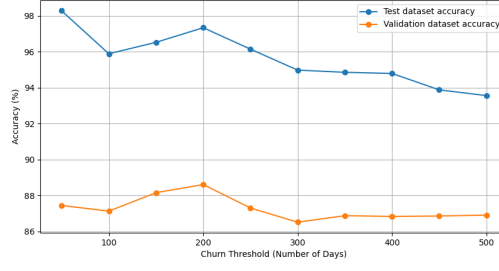


Figure 28: Logistic regression's accuracy against churn thresholds using the finance dataset

F Further Visualizations of the BG Model

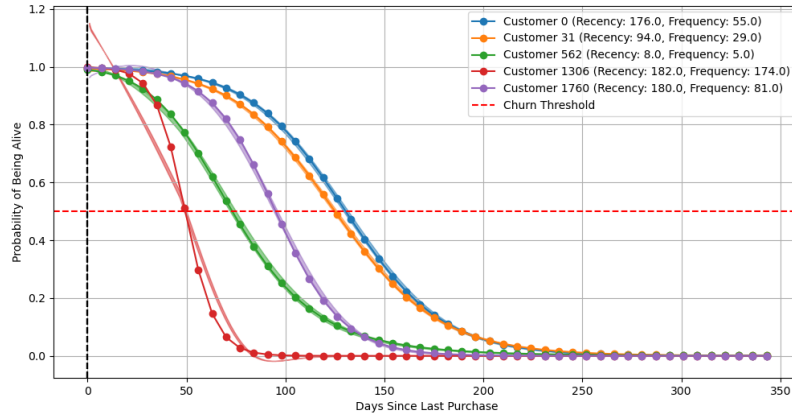


Figure 29: Plot of the mean of $P(\text{alive})$ against time for 5 different customers over the next year, using the first 5000 customers in the dataset to estimate the posterior distributions, with a churn threshold of $P(\text{alive}) = 0.5$. The shaded regions around the mean represent the HDI (High Density Interval) probability of 89% around the values of $P(\text{alive})$.

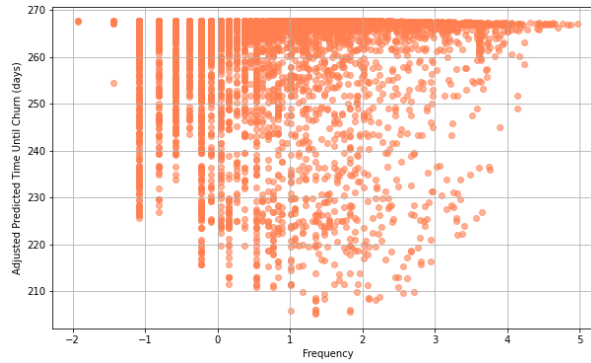


Figure 30: Predicted time until churn against frequency